KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6351A | 6 Credits: 4 | Program Elective | Source-grounded

Habitat Engineering

To introduce basic principles of town planning and the concept of vernacular architecture

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6351A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6351A.pdf from the uploaded official SITTTR syllabus archive.
Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Civil and Environmental Engineering
Course code	6351A
Course title	Habitat Engineering
Semester	6 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce basic principles of town planning and the concept of vernacular architecture
- To impart knowledge about the effects of different climatic elements which have to be
- considered before fixing the orientation of a building
- To identify the principles of landscape architecture and to enable the students to prepare a landscape layout for a site
- To outline the principles of interior design and to enable the students to design an interior space

Course outcomes

CO	Official outcome	Study level
CO1	14 Understanding describe the concept of vernacular architecture Explain the concepts of climate responsive	See official syllabus
CO2	15 Understanding architecture,lighting,ventilationandacoustics Illustrate the landscape layout for a given plot	See official syllabus
CO3	15 Applying using the principles of landscape architecture Illustrate the 3D visualization of an interior space	See official syllabus
CO4	14 Applying using the principles of interior design SeriesTests 2 CO-POMapping: Course PO1 PO2 PO3 PO4 PO5 PO6 PO7 Outcomes	See official syllabus
CO1	3 3	See official syllabus
CO2	3 3 3	See official syllabus
CO3	3 3 3	See official syllabus

CO	Official outcome	Study level
CO4	3 3 3 3-Strongly mapped,2-Moderately mapped,1-Weakly mapped	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
The	The CO-PO mapping is reproduced in the official source PDF where provided.

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	vernacular architecture	4	As specified
Module 2	ventilation and acoustics	4	As specified
Module 3	landscape architecture	4	As specified
Module 4	design	4	As specified

MODULE 1

vernacular architecture

This module is presented from the official syllabus scope for Habitat Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: vernacular architecture
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding dwelling typologies Identify the principles of town planning and the

3 Understanding dwelling typologies Identify the principles of town planning and the is an official topic in Module 1 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding dwelling typologies Identify the principles of town planning and the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding dwelling typologies identify the principles of town planning and the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding dwelling typologies Identify the principles of town planning and the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding dwelling typologies Identify the principles of town planning and the □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 4 Understanding methods for slum clearance Discuss the significance of vernacular

4 Understanding methods for slum clearance Discuss the significance of vernacular is an official topic in Module 1 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding methods for slum clearance Discuss the significance of vernacular using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding methods for slum clearance discuss the significance of vernacular should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding methods for slum clearance Discuss the significance of vernacular means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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▼ architecture and identify the characteristics of the 3 Understanding vernacular architecture of Kerala Explain the elements and principles of

architecture and identify the characteristics of the 3 Understanding vernacular architecture of Kerala Explain the elements and principles of is an official topic in Module 1 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify architecture and identify the characteristics of the 3 Understanding vernacular architecture of Kerala Explain the elements and principles of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, architecture and identify the characteristics of the 3 understanding vernacular architecture of kerala explain the elements and principles of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What architecture and identify the characteristics of the 3 Understanding vernacular architecture of Kerala Explain the elements and principles of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 1-[illegible] architecture and identify the characteristics of the 3 Understanding vernacular architecture of Kerala Explain the elements and principles of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 4 Understanding architectural design Habitat, Settlement Pattern, Dwelling/Housing Typology Town Planning- Growth of towns - Principles of town planning, Zoning, Bye-laws, Master Plan Housing-Importance of housing-Slums-Types-Causes-Characteristics-Effects- Agencies and Methods of Slum clearance, Mass Housing. Vernacular architecture-Indigenous construction materials, Vernacular architecture of Kerala, Primary elements of Architectural design - Point, Line, Plane, Volume. Form - Definition, Regular and irregular forms. Principles of composition- Balance, Rhythm, Emphasis, Proportion and scale, Movement, Contrast, Unity.Space- Exterior and Interior space Explain the concepts of climate responsive architecture, lighting,

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding architectural design Habitat, Settlement Pattern, Dwelling/Housing Typology Town Planning- Growth of towns - Principles of town planning, Zoning, Bye-laws, Master Plan Housing-Importance of housing-Slums-Types-Causes-Characteristics-Effects-Agencies and Methods of Slum clearance, Mass Housing. Vernacular architecture-Indigenous construction materials, Vernacular architecture of Kerala, Primary elements of Architectural design - Point, Line, Plane, Volume. Form - Definition, Regular and irregular forms. Principles of composition- Balance, Rhythm, Emphasis, Proportion and scale, Movement, Contrast, Unity.Space- Exterior and Interior space Explain the concepts of climate responsive architecture, lighting, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding architectural design habitat, settlement pattern, dwelling/housing typology town planning- growth of towns - principles of town planning, zoning, bye-laws, master plan housing-importance of housing-slums-types-causes-characteristics-effects- agencies and methods of slum clearance, mass housing. vernacular architecture-indigenous construction materials, vernacular architecture of kerala, primary elements of architectural design - point, line, plane, volume. form - definition, regular and irregular forms. principles of composition- balance, rhythm, emphasis, proportion and scale, movement, contrast, unity.space- exterior and interior space explain the concepts of climate responsive architecture, lighting, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding architectural design Habitat, Settlement Pattern, Dwelling/Housing Typology Town Planning- Growth of towns - Principles of town planning, Zoning, Bye-laws, Master Plan Housing-Importance of housing-Slums-Types-Causes-Characteristics-Effects- Agencies and Methods of Slum clearance, Mass Housing. Vernacular architecture- Indigenous construction materials, Vernacular architecture of Kerala, Primary elements of Architectural design - Point, Line, Plane, Volume. Form - Definition, Regular and irregular forms. Principles of composition- Balance, Rhythm, Emphasis, Proportion and scale, Movement, Contrast, Unity.Space- Exterior and Interior space Explain the concepts of climate responsive architecture, lighting, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

ventilation and acoustics

This module is presented from the official syllabus scope for Habitat Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: ventilation and acoustics
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding elements of climate Discuss the use of sun path diagrams and wind

4 Understanding elements of climate Discuss the use of sun path diagrams and wind is an official topic in Module 2 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding elements of climate Discuss the use of sun path diagrams and wind using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding elements of climate discuss the use of sun path diagrams and wind should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding elements of climate Discuss the use of sun path diagrams and wind means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding elements of climate Discuss the use of sun path diagrams and wind □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 3 Understanding rose diagrams to fix the orientation of a building

3 Understanding rose diagrams to fix the orientation of a building is an official topic in Module 2 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding rose diagrams to fix the orientation of a building using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding rose diagrams to fix the orientation of a building should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding rose diagrams to fix the orientation of a building means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 2-□□ 3 Understanding rose diagrams to fix the orientation of a building □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain the indices of thermal comfort 4 Understanding Describe the concepts of day lighting, ventilation

Explain the indices of thermal comfort 4 Understanding Describe the concepts of day lighting, ventilation is an official topic in Module 2 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the indices of thermal comfort 4 Understanding Describe the concepts of day lighting, ventilation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the indices of thermal comfort 4 understanding describe the concepts of day lighting, ventilation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the indices of thermal comfort 4 Understanding Describe the concepts of day lighting, ventilation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain the indices of thermal comfort 4 Understanding Describe the concepts of day lighting, ventilation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 4 Understanding and acoustics SeriesTest-I 1 Climatology- Climate and Weather - Koppen classification – Macro Climate, Site Climate/ Micro Climate: - Local factors - temperature and humidity - Air movement - Vegetation - Urban Heat Island Elements of climate - temperature, humidity, precipitation, radiation, wind- Climate zones in India - Tropical Climate - Climatic design recommendations for tropical climate Sun path diagram/Solarchart - Global wind pattern - Wind movement - Wind rosediagram. Human comfort in buildings -Thermal comfort - Indices of thermal comfort -Reduction of solar heat gain-Sol-air temperature Lighting of interior and exterior - measurement of light - Glare - Day lighting - Sky component - SP 41 Natural ventilation - Artificial ventilation- Functions of Ventilation - Factors affecting indoor airflow - airflow around buildings - Openings-Cross ventilation - Passive cooling Acoustics-sources-Soundandnoise-Noisecontrol. Illustrate the landscape layout for a given plot using the principles of

4 Understanding and acoustics SeriesTest-I 1 Climatology- Climate and Weather - Koppen classification – Macro Climate, Site Climate/ Micro Climate: - Local factors - temperature and humidity - Air movement - Vegetation - Urban Heat Island Elements of climate - temperature, humidity, precipitation, radiation, wind- Climate zones in India - Tropical Climate - Climatic design recommendations for tropical climate Sun path diagram/Solarchart - Global wind pattern - Wind movement - Wind rosediagram. Human comfort in buildings -Thermal comfort - Indices of thermal comfort -Reduction of solar heat gain-Sol-air temperature Lighting of interior and exterior - measurement of light - Glare - Day lighting - Sky component - SP 41 Natural ventilation - Artificial ventilation- Functions of Ventilation - Factors affecting indoor airflow - airflow around buildings - Openings-Cross ventilation - Passive cooling Acoustics-sources-Soundandnoise-Noisecontrol. Illustrate the landscape layout for a given plot using the principles of is an official topic in Module 2 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding and acoustics SeriesTest-I 1 Climatology-Climate and Weather - Koppen classification – Macro Climate, Site Climate/ Micro Climate: - Local factors - temperature and humidity - Air movement - Vegetation - Urban Heat Island Elements of climate - temperature, humidity, precipitation, radiation, wind- Climate zones in India - Tropical Climate - Climatic design recommendations for tropical climate Sun path diagram/Solarchart - Global wind pattern - Wind movement - Wind rosediagram. Human comfort in buildings -Thermal comfort - Indices of thermal comfort -Reduction of solar heat gain-Sol-air temperature Lighting of interior and exterior - measurement of light - Glare - Day lighting - Sky component - SP 41 Natural ventilation - Artificial ventilation- Functions of Ventilation - Factors affecting indoor airflow - airflow around buildings - Openings-Cross ventilation - Passive cooling Acoustics-sources-Soundandnoise-Noisecontrol. Illustrate the landscape layout for a given plot using the principles of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding and acoustics seriestest-i 1 climatology-climate and weather - koppen classification – macro climate, site climate/ micro climate: - local factors - temperature and humidity - air movement - vegetation - urban heat island elements of climate - temperature, humidity, precipitation, radiation, wind- climate zones in india - tropical climate - climatic design recommendations for tropical climate sun path diagram/solarchart - global wind pattern - wind movement - wind rosediagram. human comfort in buildings - thermal comfort - indices of thermal comfort -reduction of solar heat gain-sol-air temperature lighting of interior and exterior - measurement of light - glare - day lighting - sky component - sp 41 natural ventilation - artificial ventilation-functions of ventilation - factors affecting indoor airflow - airflow around buildings - openings-cross ventilation - passive cooling acoustics-sources-soundandnoise-noisecontrol. illustrate the landscape layout for a given plot using the principles of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding and acoustics SeriesTest-I 1 Climatology- Climate and Weather - Koppen classification – Macro Climate, Site Climate/ Micro Climate: - Local factors - temperature and humidity - Air movement - Vegetation - Urban Heat Island Elements of climate - temperature, humidity, precipitation, radiation, wind- Climate zones in India - Tropical Climate - Climatic design recommendations for tropical climate Sun path diagram/Solarchart - Global wind pattern - Wind movement - Wind rosediagram. Human comfort in buildings -Thermal comfort - Indices of thermal comfort -Reduction of solar heat gain-Sol-air temperature Lighting of interior and exterior - measurement of light - Glare - Day lighting - Sky component - SP 41 Natural ventilation - Artificial ventilation- Functions of Ventilation - Factors affecting indoor airflow - airflow around buildings - Openings-Cross ventilation - Passive cooling Acoustics-sources-Soundandnoise-Noisecontrol. Illustrate the landscape layout for a given plot using the principles of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding and acoustics SeriesTest-I 1 Climatology- Climate and Weather - Koppen classification – Macro Climate, Site Climate/ Micro Climate: - Local factors - temperature and humidity - Air movement - Vegetation - Urban Heat Island Elements of climate - temperature, humidity, precipitation, radiation, wind- Climate zones in India - Tropical Climate - Climatic design recommendations for tropical climate Sun path diagram/Solarchart - Global wind pattern - Wind movement - Wind rosediagram. Human comfort in buildings -Thermal comfort - Indices of thermal comfort -Reduction of solar heat gain-Sol-air temperature Lighting of interior and exterior - measurement of light - Glare - Day lighting - Sky component - SP 41 Natural ventilation - Artificial ventilation- Functions of Ventilation - Factors affecting indoor airflow - airflow around buildings - Openings-Cross ventilation - Passive cooling Acoustics-sources-Soundandnoise-Noisecontrol. Illustrate the landscape layout for a given plot using the principles of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

landscape architecture

This module is presented from the official syllabus scope for Habitat Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: landscape architecture
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding phytogeographical regions of India Explain the principles and elements of landscape

4 Understanding phytogeographical regions of India Explain the principles and elements of landscape is an official topic in Module 3 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding phytogeographical regions of India Explain the principles and elements of landscape using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding phytogeographical regions of india explain the principles and elements of landscape should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding phytogeographical regions of India Explain the principles and elements of landscape means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 4 Understanding phytogeographical regions of India Explain the principles and elements of landscape [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 3 Understanding design Explain the elements and features of Mughal,

3 Understanding design Explain the elements and features of Mughal, is an official topic in Module 3 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding design Explain the elements and features of Mughal, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding design explain the elements and features of mughal, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding design Explain the elements and features of Mughal, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Understanding design Explain the elements and features of Mughal, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 4 Understanding French, Japanese and English gardens

4 Understanding French, Japanese and English gardens is an official topic in Module 3 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding French, Japanese and English gardens using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding french, japanese and english gardens should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding French, Japanese and English gardens means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding French, Japanese and English gardens □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▼ Sketch landscape layout for a given site 4 Applying Landscape Architecture: Phytogeographical regions of India Principles - Unity, Repetition, Order, Proportion, Elements - Line, Form, Texture, Colour Visual weight Components of Landscape design - Softscape - Classification of Plants, Decorative vegetation, Species typical to Kerala, Lawn, Turf -Hardscape - Ferrocement, Paving blocks, Stone items, Artificially moulded elements, walkways, patios, driveways, walls, Site furniture etc. - Water as an element Formal and Informal Gardens - Elements and features of Gardens - Mughal Gardens - English Gardens - French Gardens - Japanese Gardens (Sketch layout). Traditional Home gardens of Kerala Bubble diagram and concept sketches Illustrate the 3Dvisualization of a space using the principles of interior

Sketch landscape layout for a given site 4 Applying Landscape Architecture: Phytogeographical regions of India Principles - Unity, Repetition, Order, Proportion, Elements - Line, Form, Texture, Colour Visual weight Components of Landscape design - Softscape - Classification of Plants, Decorative vegetation, Species typical to Kerala, Lawn, Turf - Hardscape - Ferrocement, Paving blocks, Stone items, Artificially moulded elements, walkways, patios, driveways, walls, Site furniture etc. - Water as an element Formal and Informal Gardens - Elements and features of Gardens - Mughal Gardens - English Gardens - French Gardens - Japanese Gardens (Sketch layout). Traditional Home gardens of Kerala Bubble diagram and concept sketches Illustrate the 3Dvisualization of a space using the principles of interior is an official topic in Module 3 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Sketch landscape layout for a given site 4 Applying Landscape Architecture: Phytogeographical regions of India Principles - Unity, Repetition, Order, Proportion, Elements - Line, Form, Texture, Colour Visual weight Components of Landscape design - Softscape - Classification of Plants, Decorative vegetation, Species typical to Kerala, Lawn, Turf - Hardscape - Ferrocement, Paving blocks, Stone items, Artificially moulded elements, walkways, patios, driveways, walls, Site furniture etc. - Water as an element Formal and Informal Gardens - Elements and features of Gardens - Mughal Gardens - English Gardens - French Gardens - Japanese Gardens (Sketch layout). Traditional Home gardens of Kerala Bubble diagram and concept sketches Illustrate the 3Dvisualization of a space using the principles of interior using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sketch landscape layout for a given site 4 applying landscape architecture: phytogeographical regions of india principles - unity, repetition, order, proportion, elements - line, form, texture, colour visual weight components of landscape design - softscape - classification of plants, decorative vegetation, species typical to kerala, lawn, turf -hardscape - ferrocement, paving blocks, stone items, artificially moulded elements, walkways, patios, driveways, walls, site furniture etc. - water as an element formal and informal gardens - elements and features of gardens - mughal gardens - english gardens - french gardens - japanese gardens (sketch layout). traditional home gardens of kerala bubble diagram and concept sketches illustrate the 3dvisualization of a space using the principles of interior should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Sketch landscape layout for a given site 4 Applying Landscape Architecture: Phytogeographical regions of India Principles - Unity, Repetition, Order, Proportion, Elements - Line, Form, Texture, Colour Visual weight Components of Landscape design - Softscape - Classification of Plants, Decorative vegetation, Species typical to Kerala, Lawn, Turf -Hardscape - Ferrocement, Paving blocks, Stone items, Artificially moulded elements, walkways, patios, driveways, walls, Site furniture etc. - Water as an element Formal and Informal Gardens - Elements and features of Gardens - Mughal Gardens - English Gardens - French Gardens - Japanese Gardens (Sketch layout). Traditional Home gardens of Kerala Bubble diagram and concept sketches Illustrate the 3Dvisualization of a space using the principles of interior means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Sketch landscape layout for a given site 4 Applying Landscape Architecture: Phytogeographical regions of India Principles - Unity, Repetition, Order, Proportion, Elements - Line, Form, Texture, Colour Visual weight Components of Landscape design - Softscape - Classification of Plants, Decorative vegetation, Species typical to Kerala, Lawn, Turf -Hardscape - Ferrocement, Paving blocks, Stone items, Artificially moulded elements, walkways, patios, driveways, walls, Site furniture etc. - Water as an element Formal and Informal Gardens - Elements and features of Gardens - Mughal Gardens - English Gardens - French Gardens - Japanese Gardens (Sketch layout). Traditional Home gardens of Kerala Bubble diagram and concept sketches Illustrate the 3Dvisualization of a space using the principles of interior □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▶ Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

design

This module is presented from the official syllabus scope for Habitat Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: design
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding Design

3 Understanding Design is an official topic in Module 4 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Design using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding design should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Design means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding Design □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Discuss the techniques to modify interiors 4 Understanding Familiarize with the anthropometric data sheets

Discuss the techniques to modify interiors 4 Understanding Familiarize with the anthropometric data sheets is an official topic in Module 4 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the techniques to modify interiors 4 Understanding Familiarize with the anthropometric data sheets using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the techniques to modify interiors 4 understanding familiarize with the anthropometric data sheets should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Discuss the techniques to modify interiors 4 Understanding Familiarize with the anthropometric data sheets means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Discuss the techniques to modify interiors 4 Understanding Familiarize with the anthropometric data sheets □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

▼ 3 Understanding for interior design

3 Understanding for interior design is an official topic in Module 4 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding for interior design using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding for interior design should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding for interior design means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding for interior design □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Illustrate 3Dvisualization of an interior space 4 Applying SeriesTest-II 1 Interior Space - Positive and Negative space - Circulation space - Physical quantities and Attached qualities (Tactile, Thermal, Auditory and Olfactory) Elements of Interior design - Point, Line, Plane, Form, Shape, Texture, Colour (Colour wheel, Munsell Scale) and Light Principles of Interior Design - Proportion, Scale, Balance, Harmony, Unity, Variety, Rhythm, Emphasis Modification of interiors – Space modulation to modify interiors – Structural and Non- structural modulation - Artificial Lighting - Cool and Warm lighting Classic and modern decorative materials Furniture-Classification based on styles, materials and function – Modular furniture Finishes - for walls, floors and ceilings - criteria for selection Fabric-curtains,blinds,carpets,rugs Ergonomics and Anthropometry - Activity relationships - Planning and Three Dimensional Visualization of interiors Text/Reference: T/R BookTitle/Author T1 FrancisDKChing-ArchitectureForm,Space,Order;JohnWiley&Sons,Inc Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman R1 MartinEvans:Housing,ClimateandComfort;TheArchitecturalPress, London R2 R3 Town Planning; Rangwala, Charotar Publication R4 Givoni-Man,Climate and Architecture, Applied Science Publishers TimeSaverStandardsforLandscapeDesign,CharlesW.HarrisandNicholas T R5 Dines MichaelLaurie-AnIntroductiontolandscapearchitecture,AmericanElsevier R6 Publishing Company TimeSaverStandardsforInteriorDesignandSpace,JosephDeChiara,Julius Panero, R7 and Martin Zelnik, McGraw-Hill, Inc OnlineResources: Sl.No WebsiteLink 1 https://ncert.nic.in/ncerts/l/legy110.pdf 2 https://www.yourhome.gov.au/passive-design 3 https://en.wikipedia.org/

Illustrate 3Dvisualization of an interior space 4 Applying SeriesTest-II 1 Interior Space - Positive and Negative space - Circulation space - Physical quantities and Attached qualities (Tactile, Thermal, Auditory and Olfactory) Elements of Interior design - Point, Line, Plane, Form, Shape, Texture, Colour (Colour wheel, Munsell Scale) and Light Principles of Interior Design

- Proportion, Scale, Balance, Harmony, Unity, Variety, Rhythm, Emphasis Modification of interiors – Space modulation to modify interiors – Structural and Non- structural modulation - Artificial Lighting - Cool and Warm lighting Classic and modern decorative materials Furniture-Classification based on styles, materials and function – Modular furniture Finishes - for walls, floors and ceilings - criteria for selection Fabric-curtains,blinds,carpets,rugs Ergonomics and Anthropometry - Activity relationships - Planning and Three Dimensional Visualization of interiors Text/Reference: T/R BookTitle/Author T1 FrancisDKChing-ArchitectureForm,Space,Order;JohnWiley&Sons,Inc Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman R1 MartinEvans:Housing,ClimateandComfort;TheArchitecturalPress, London R2 R3 Town Planning; Rangwala, Charotar Publication R4 Givoni-Man,Climate and Architecture, Applied Science Publishers TimeSaverStandardsforLandscapeDesign,CharlesW.HarrisandNicholas T R5 Dines MichaelLaurie-AnIntroductiontolandscapearchitecture,AmericanElsevier R6 Publishing Company TimeSaverStandardsforInteriorDesignandSpace,JosephDeChiara,Julius Panero, R7 and Martin Zelnik, McGraw-Hill, Inc OnlineResources: Sl.No WebsiteLink 1 https://ncert.nic.in/ncerts/l/legy110.pdf 2 https://www.yourhome.gov.au/passive-design 3 https://en.wikipedia.org/ is an official topic in Module 4 of Habitat Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate 3Dvisualization of an interior space 4 Applying SeriesTest-II 1 Interior Space - Positive and Negative space - Circulation space - Physical quantities and Attached qualities (Tactile, Thermal, Auditory and Olfactory) Elements of Interior design - Point, Line, Plane, Form, Shape, Texture, Colour (Colour wheel, Munsell Scale) and Light Principles of Interior Design - Proportion, Scale, Balance, Harmony, Unity, Variety, Rhythm, Emphasis Modification of interiors – Space modulation to modify interiors – Structural and Non- structural modulation - Artificial Lighting - Cool and Warm lighting Classic and modern decorative materials Furniture- Classification based on styles, materials and function – Modular furniture Finishes - for walls, floors and ceilings - criteria for selection Fabric- curtains,blinds,carpets,rugs Ergonomics and Anthropometry - Activity relationships - Planning and Three Dimensional Visualization of interiors Text/Reference: T/R BookTitle/Author T1 FrancisDKChing-ArchitectureForm,Space,Order;JohnWiley&Sons,Inc Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman R1 MartinEvans:Housing,ClimateandComfort;TheArchitecturalPress, London R2 R3 Town Planning; Rangwala, Charotar Publication R4 Givoni-Man,Climate and Architecture, Applied Science Publishers TimeSaverStandardsforLandscapeDesign,CharlesW.HarrisandNicholas T R5 Dines MichaelLaurie-AnIntroductiontolandscapearchitecture,AmericanElsevier R6 Publishing Company TimeSaverStandardsforInteriorDesignandSpace,JosephDeChiara,Julius Panero, R7 and Martin Zelnik, McGraw-Hill, Inc OnlineResources: Sl.No WebsiteLink 1 https://ncert.nic.in/ncerts/l/legy110.pdf 2 https://www.yourhome.gov.au/passive-design 3 https://en.wikipedia.org/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate 3dvisualization of an interior space 4 applying seriestest-ii 1 interior space - positive and negative space - circulation space - physical quantities and attached qualities (tactile, thermal, auditory and olfactory) elements of interior design - point, line, plane, form, shape, texture, colour (colour wheel, munsell scale) and light principles of interior design - proportion, scale, balance, harmony, unity, variety, rhythm, emphasis modification of interiors – space modulation to modify interiors – structural and non- structural modulation - artificial lighting - cool and warm lighting classic and modern decorative materials furniture-classification based on styles, materials and function – modular furniture finishes - for walls, floors and ceilings - criteria for selection fabric-curtains,blinds,carpets,rugs ergonomics and anthropometry - activity relationships - planning and three dimensional visualization of interiors text/reference: t/r booktitle/author t1 francisdkching-architectureform,space,order;johnwiley&sons,inc koenigsbergeret-al:manual of tropical housing and building; orient longman r1 martinevans:housing,climateandcomfort;thearchitecturalpress, london r2 r3 town planning; rangwala, charotar publication r4 givoni-man,climate and architecture, applied science publishers timesaverstandardsforlandscapedesign,charlesw.harrisandnicholas t r5 dines michaellaurie-anintroductiontolandscapearchitecture,americanelsevier r6 publishing company timesaverstandardsforinteriordesignandspace,josephdechiara,julius panero, r7 and martin zelnik, mcgraw-hill, inc onlineresources: sl.no websitelink 1 https://ncert.nic.in/ncerts/l/legy110.pdf 2 https://www.yourhome.gov.au/passive-design 3 https://en.wikipedia.org/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate 3Dvisualization of an interior space 4 Applying SeriesTest-II 1 Interior Space - Positive and Negative space - Circulation space - Physical quantities and Attached qualities (Tactile, Thermal, Auditory and Olfactory) Elements of Interior design - Point, Line, Plane, Form, Shape, Texture, Colour (Colour wheel, Munsell Scale) and Light Principles of Interior Design - Proportion, Scale, Balance, Harmony, Unity, Variety, Rhythm, Emphasis Modification of interiors – Space

Aspect	What to prepare
	modulation to modify interiors – Structural and Non- structural modulation - Artificial Lighting - Cool and Warm lighting Classic and modern decorative materials Furniture-Classification based on styles, materials and function – Modular furniture Finishes - for walls, floors and ceilings - criteria for selection Fabric-curtains,blinds,carpets,rugs Ergonomics and Anthropometry - Activity relationships - Planning and Three Dimensional Visualization of interiors Text/Reference: T/R BookTitle/Author T1 FrancisDKChing-ArchitectureForm,Space,Order;JohnWiley&Sons,Inc Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman R1 MartinEvans:Housing,ClimateandComfort;TheArchitecturalPress, London R2 R3 Town Planning; Rangwala, Charotar Publication R4 Givoni-Man,Climate and Architecture, Applied Science Publishers TimeSaverStandardsforLandscapeDesign,CharlesW.HarrisandNicholas T R5 Dines MichaelLaurie-AnIntroductiontolandscapearchitecture,AmericanElsevier R6 Publishing Company TimeSaverStandardsforInteriorDesignandSpace,JosephDeChiara,Julius Panero, R7 and Martin Zelnik, McGraw-Hill, Inc OnlineResources: Sl.No WebsiteLink 1 https://ncert.nic.in/ncerts/l/legy110.pdf 2 https://www.yourhome.gov.au/passive-design 3 https://en.wikipedia.org/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate 3Dvisualization of an interior space 4 Applying SeriesTest-II 1 Interior Space - Positive and Negative space - Circulation space - Physical quantities and Attached qualities (Tactile, Thermal, Auditory and Olfactory) Elements of Interior design - Point, Line, Plane, Form, Shape, Texture, Colour (Colour wheel, Munsell Scale) and Light Principles of Interior Design - Proportion, Scale, Balance, Harmony, Unity, Variety, Rhythm, Emphasis Modification of interiors – Space modulation to modify interiors – Structural and Non- structural modulation - Artificial Lighting - Cool and Warm lighting Classic and modern decorative materials Furniture-Classification based on styles, materials and function – Modular furniture Finishes - for walls, floors and ceilings - criteria for selection Fabric-curtains,blinds,carpets,rugs Ergonomics and Anthropometry - Activity relationships - Planning and Three Dimensional Visualization of interiors Text/Reference: T/R BookTitle/Author T1 FrancisDKChing-ArchitectureForm,Space,Order;JohnWiley&Sons,Inc Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman R1 MartinEvans:Housing,ClimateandComfort;TheArchitecturalPress, London R2 R3 Town Planning; Rangwala, Charotar Publication R4 Givoni-Man,Climate and Architecture, Applied Science Publishers TimeSaverStandardsforLandscapeDesign,CharlesW.HarrisandNicholas T R5 Dines MichaelLaurie-AnIntroductiontolandscapearchitecture,AmericanElsevier R6 Publishing Company TimeSaverStandardsforInteriorDesignandSpace,JosephDeChiara,Julius Panero, R7 and Martin Zelnik, McGraw-Hill, Inc OnlineResources: Sl.No WebsiteLink 1 https://ncert.nic.in/ncerts/l/legy110.pdf 2 https://www.yourhome.gov.au/passive-design 3 https://en.wikipedia.org/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 3 Understanding dwelling typologies Identify the principles of town planning and the. (3 marks)**

▶ **Define or explain 4 Understanding methods for slum clearance Discuss the significance of vernacular. (3 marks)**

▶ **Define or explain architecture and identify the characteristics of the 3 Understanding vernacular architecture of Kerala Explain the elements and principles of. (3 marks)**

▶ **Define or explain 4 Understanding architectural design Habitat, Settlement Pattern, Dwelling/Housing Typology Town Planning- Growth of towns - Principles of town planning, Zoning, Bye-laws, Master Plan Housing-Importance of housing-Slums-Types-Causes-Characteristics-Effects- Agencies and Methods of Slum clearance, Mass Housing. Vernacular architecture-Indigenous construction materials, Vernacular architecture of Kerala, Primary elements of Architectural design - Point, Line, Plane, Volume. Form - Definition, Regular and irregular forms. Principles of composition- Balance, Rhythm, Emphasis, Proportion and scale, Movement, Contrast, Unity.Space- Exterior and Interior space Explain the concepts of climate responsive architecture, lighting,. (3 marks)**

Module 2

▶ **Define or explain 4 Understanding elements of climate Discuss the use of sun path diagrams and wind. (3 marks)**

▶ **Define or explain 3 Understanding rose diagrams to fix the orientation of a building. (3 marks)**

▶ **Define or explain Explain the indices of thermal comfort 4 Understanding Describe the concepts of day lighting, ventilation. (3 marks)**

▶ **Define or explain 4 Understanding and acoustics SeriesTest-I 1 Climatology- Climate and Weather - Koppen classification – Macro Climate, Site Climate/ Micro Climate: - Local factors - temperature and humidity - Air movement - Vegetation - Urban Heat Island Elements of climate - temperature, humidity, precipitation, radiation, wind- Climate zones in India - Tropical Climate - Climatic design recommendations for tropical climate Sun path diagram/Solarchart - Global wind pattern - Wind movement - Wind rosediagram. Human comfort in buildings - Thermal comfort - Indices of thermal comfort -Reduction of solar heat gain-Sol-air temperature Lighting of interior and exterior - measurement of light - Glare - Day lighting - Sky component - SP 41 Natural ventilation - Artificial ventilation- Functions of Ventilation - Factors affecting indoor airflow - airflow around buildings - Openings- Cross ventilation - Passive cooling Acoustics-sources-Soundandnoise-Noisecontrol. Illustrate the landscape layout for a given plot using the principles of. (3 marks)**

Module 3

▶ **Define or explain 4 Understanding phytogeographical regions of India Explain the principles and elements of landscape. (3 marks)**

▶ **Define or explain 3 Understanding design Explain the elements and features of Mughal,. (3 marks)**

▶ **Define or explain 4 Understanding French, Japanese and English gardens. (3 marks)**

▶ **Define or explain Sketch landscape layout for a given site 4 Applying Landscape Architecture: Phytogeographical regions of India Principles - Unity, Repetition, Order, Proportion, Elements - Line, Form, Texture, Colour Visual weight Components of Landscape design - Softscape - Classification of Plants, Decorative vegetation, Species typical to Kerala, Lawn, Turf -Hardscape - Ferrocement, Paving blocks, Stone items, Artificially moulded elements, walkways, patios, driveways, walls, Site furniture etc. - Water as an element Formal and Informal Gardens - Elements and features of Gardens - Mughal Gardens - English Gardens - French Gardens - Japanese Gardens (Sketch layout). Traditional Home gardens of Kerala Bubble diagram and concept sketches Illustrate the 3Dvisualization of a space using the principles of interior. (3 marks)**

Module 4

▶ **Define or explain 3 Understanding Design. (3 marks)**

▶ **Define or explain Discuss the techniques to modify interiors 4 Understanding Familiarize with the anthropometric data sheets. (3 marks)**

▶ **Define or explain 3 Understanding for interior design. (3 marks)**

▶ **Define or explain Illustrate 3Dvisualization of an interior space 4 Applying SeriesTest-II 1 Interior Space - Positive and Negative space - Circulation space - Physical quantities and Attached qualities (Tactile, Thermal, Auditory and Olfactory) Elements of Interior design - Point, Line, Plane, Form, Shape, Texture, Colour (Colour wheel, Munsell Scale) and Light Principles of Interior Design - Proportion, Scale, Balance, Harmony, Unity, Variety, Rhythm, Emphasis Modification of interiors –**

Space modulation to modify interiors – Structural and Non- structural modulation - Artificial Lighting - Cool and Warm lighting Classic and modern decorative materials Furniture-Classification based on styles, materials and function – Modular furniture Finishes - for walls, floors and ceilings - criteria for selection Fabric-curtains,blinds,carpets,rugs Ergonomics and Anthropometry - Activity relationships - Planning and Three Dimensional Visualization of interiors Text/Reference: T/R BookTitle/Author T1 FrancisDKChing-ArchitectureForm,Space,Order;JohnWiley&Sons,Inc Koenigsbergeret-al:Manual of tropical housing and building; Orient Longman R1 MartinEvans:Housing,ClimateandComfort;TheArchitecturalPress, London R2 R3 Town Planning; Rangwala, Charotar Publication R4 Givoni-Man,Climate and Architecture, Applied Science Publishers TimeSaverStandardsforLandscapeDesign,CharlesW.HarrisandNicholas T R5 Dines MichaelLaurie-AnIntroductiontolandscapearchitecture,AmericanElsevier R6 Publishing Company TimeSaverStandardsforInteriorDesignandSpace,JosephDeChiara,Julius Panero, R7 and Martin Zelnik, McGraw-Hill, Inc OnlineResources: Sl.No WebsiteLink 1 https://ncert.nic.in/ncerts/l/legy110.pdf 2 https://www.yourhome.gov.au/passive-design 3 https://en.wikipedia.org/. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding dwelling typologies Identify the principles of town planning and the**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding elements of climate Discuss the use of sun path diagrams and wind**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Understanding phytogeographical regions of India Explain the principles and elements of landscape**.

Module 4

1-mark definition: State the meaning of **3 Understanding Design**.

3-mark explanation: Explain one principle, process or comparison from this module.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 6351A. Verify institutional updates against the current official curriculum.